Supplementary Material

Evaluating Reasoning Strategies for Interactive Transcriptomic Interpretation

Muiz Khan, Alan Carbajo, Cristiana Iosef, Suzan Arslanturk, and Sorin Draghici

This supplement supports the statements of the paper with the underlying tables, figures, and benchmark examples. Section and figure references without the prefix S refer to the paper. All scores are macro-averaged over the six datasets and graded by Claude Opus 5.5 unless stated otherwise; data, prompts, and the scripts that produce every table and figure are available at `https://github.com/MuizZer07/Nova-Reasoning-Workflows`.

Contents. Datasets: Table S1. Benchmark examples: Tables S2–S5. Nova system prompt: Fig. S1. RQ1 (reasoning structure and cost): Tables S6–S8 and Fig. S2. RQ2 (frontier model): Table S9 and Fig. S3. RQ3 (questions to decisions): Figs. S4 and S5. RQ4 (robustness): Tables S10–S12 and Fig. S6. Worked example: Fig. S7.

Table S1. iPathwayGuide [1] analysis summary of the six datasets (counts before multiple-testing correction). Paper, Sec. 3.1 and Fig. 1B.

Dataset	Category	Genes	DEGs	GO terms	Pathways	Upstream miRNAs	Upstream genes	Diseases
GSE47363 [2]	Tumor supp.	20,526	1,166	1,974	113	118	194	27
GSE147507 [3]	Infection	9,400	812	1,040	38	2	74	125
GSE150316 [4]	Infection	15,112	229	544	15	39	110	236
GSE26487 [5]	Drug resp.	8,577	62	619	19	0	140	126
GSE74000 [6]	Drug resp.	19,826	557	1,115	62	83	238	134
GSE58434 [7]	Drug resp.	13,476	699	2,212	68	0	131	200

Tables S2–S4 show one benchmark item from each setting (dataset GSE47363, miR-542-3p in U2OS osteosarcoma cells) with its reference annotations, and Table S5 shows two hypothesis steps whose qualifying evidence every system read as contradicting (paper, RQ3 and Discussion).

Table S2. An independent question (S1) and its reference annotations.

Item	NOVA-S1-D1-033; multiple choice; RRL L2; difficulty hard; LLM-assisted
Question	Which regulator is significant by target over-representation yet has an undetermined predicted direction (equal activating and inhibiting targets) and an essentially unchanged own transcript? A. PAK2; B. TP53; C. SREBF2; D. BDNF
Reference answer	A. PAK2 is significant by over-representation (FDR $\approx 2.1 \times 10^{-3}$) but has a zero activation z-score (3 activating vs. 3 inhibiting DE targets), so its direction is undetermined, and its own transcript is unchanged (logFC ≈ 0.11). TP53 and SREBF2 have strong directional calls, and BDNF is an activated regulator with a changed transcript.
Evidence layer	Upstream regulators
Provenance	`upstream_gene.tsv` [symbol = PAK2: zscore, pv_fdr]
Supporting evidence	PAK2 FDR 2.1×10^{-3}, z-score 0 (3 up / 3 down targets)
Expected conclusion	PAK2 is the significant but directionless regulator.

Table S3. A multi-turn conversation (S2, NOVA-S2-D1-T03): turns build on earlier turns (not only the preceding one), and turn 5 is a challenge turn at which the reference annotations expect a revision. The paper's publication centres on RPS23 repression; the conversation tests whether the data show a broader ribosome-biogenesis program.

Turn	Question (shortened)	Depends on	RRL
1	Beyond RPS23, do ribosomal-protein transcripts fall uniformly or selectively?	–	L1
2	Do the Ribosome and Ribosome-biogenesis pathways support a coordinated signature?	1	L2
3	Do rRNA-processing, small-subunit-biogenesis, or nucleolus GO terms come through?	1, 2	L2
4	Some ribosomal-protein genes are induced: what does RPS27L induction indicate?	1	L2
5†	Should the idea of a broad ribosome-biogenesis / nucleolar-stress program be kept as stated, or revised?	1–4	L4
6*	Do the RNA-exosome components show an inferred change pointing to the rRNA-maturation defect?	5	L3
7*	How does the revised picture sit against the publication's evidence of nucleolar stress?	3, 5	L4
8	Is RPS23 repression plausibly a direct miR-542-3p effect? What does the miRNA layer say?	1, 7	L3
9	How does the RPL11 transcript behave, and what does it say about post-transcriptional control?	7, 8	L3
10	If RPS23 were restored, what would the data-grounded prediction be, and how would it be tested?	5, 8	L3
11*	State the final, revised account of the ribosomal arm with the confidence on each part.	1, 5, 7–9	L4

*Challenge turn. †Challenge turn with an expected revision: from a "broad ribosome-biogenesis / nucleolar-stress transcriptional program" to a "selective repression of a specific ribosomal-protein subset (RPS23/RPS28/RPL22-led) with no coordinated transcriptional biogenesis signature"; the answer must integrate turns 1–4 without contradicting the selective-repression finding. All seven systems revised at turn 5 (credit 1), e.g. Nova-Thinking: "REVISE. The original claim … should be replaced with a narrower … model: miR-542-3p selectively suppresses a subset of ribosomal-protein transcripts (not the family)"; GPT-6 Instant: "It needs revision. The transcriptome supports selective repression of particular ribosomal proteins, especially RPS23/RPS28/RPL22, but not a broad ribosome-biogenesis or rRNA-processing transcriptional program."

Table S4. A hypothesis evaluation (S3, NOVA-S3-D1-H02): *p53 activation is driven solely by direct miR-542-3p repression of RPS23, which raises RPL11, promotes RPL11–MDM2 association, and thereby stabilizes p53.* Each step tests one piece of evidence; the reference annotations give its evidence relation and the hypothesis state after it. Right: the states the systems declared in their status lines (✓ matches the reference).

		Reference		Declared state		
Step	**Evidence tested**	**Relation**	**State after**	**Thinking**	**Deep**	**GPT-6 Med.**
1	Expression of the candidate ribosomal-protein transcripts	supports	partial	✓	weakened	✓
2	miR-542-3p target enrichment; the publication's direct-target evidence for RPS23	supports	partial	✓	✓	✓
3	RPL11 transcript versus the proposed translational induction of RPL11	qualifies	partial	✓	✓	weakened
4	Enrichment of ribosome / ribosome-biogenesis processes	qualifies	partial	✓	weakened	weakened
5	Publication knockdowns distinguishing RPS23 from RPS28/RPL22	supports	partial	✓	weakened	insufficient
6	Publication rescue experiment and magnitude comparison	contradicts	weakened	✓	✓	✓
7	Data-level candidates for contributions beyond RPS23	qualifies	weakened	✓	✓	✓
8	Synthesis of steps 1–7	contradicts	weakened	✓	✓	partial
Final	Decision	**revise**; weakened		✓; ✓	✓; ✓	✓; partial

Reference revised hypothesis: miR-542-3p repression of RPS23 is a *major but not the sole* route to p53 activation; additional miR-542-3p-dependent mechanisms contribute, as shown by incomplete rescue when the RPS23 3′UTR is removed. Calibration requirement: high confidence that the RPS23 → RPL11 → MDM2 route is genuine and central; reject the "sole route" wording; do not claim the RPL11 protein step is confirmed by these data (RPL11 mRNA is flat). All seven systems chose *revise*; the final-state column gives decision; state.

How to answer. Answer first. The first one or two sentences give the direct answer (the option, the value, the verdict, the conclusion). Evidence and caveats follow. … Exact numbers, as the tools returned them (logFC, adjusted p / FDR, z-score, counts like "48 of 55", ranks). Give the denominator with every count.

Evidence protocol. A. Named entities → direct lookup … Ranked top-N lists are for "what are the top…" questions only. B. Absence needs proof. Never say an item is "absent", "not significant", … because it did not appear in a truncated list. C. Use the layer that actually measures the claim: … A regulator's own mRNA change is a DIFFERENT question from its inferred activity … "Significant by ORA only" is not the same as "perturbed". … E. Look for the counter-evidence on purpose.

Reasoning rules. … Levels of evidence: transcript-level data cannot by itself establish protein abundance, protein stability, binding, localization, or a cellular phenotype. … Untested is not supported. … Significance vs trend: a value missing the threshold is "not significant (trend)", never "changed".

Calibration. … RETAIN only if it is supported as stated; QUALIFY when the core holds but its scope or certainty must be narrowed; REVISE when a central component is wrong but a modified claim is supported …; REJECT only when the core claim itself is contradicted; INSUFFICIENT EVIDENCE when the data cannot decide. Do not reach for "reject" when "revise" or "qualify" is accurate …

Fig. S1. Excerpt of the system prompt shared by all four Nova workflows (paper, Sec. 3.1), quoted verbatim and shortened (…). The full prompts of every workflow stage are in the repository (`prompts/`).

Table S5. Qualifying evidence read as contradicting (paper, RQ3 and Discussion): two hypothesis steps whose reference relation is *qualifies*, which all seven systems (four Nova workflows and three GPT-6 efforts) read as *contradicts*. Answers are quoted verbatim and shortened.

SARS-CoV-2 host response (GSE147507), NOVA-S3-D2-H03-S04	
Hypothesis	An IL-1/IL-6 inflammatory cytokine axis is the principal driver of the transcriptional signature.
Step	Treating IL1B and TNF as candidate upstream drivers, does the target-based regulator inference confirm them as significant activated regulators of the whole signature?
Reference	Qualifies; state stays *partially supported*. IL1B (18 targets, 14 consistent, z 1.79, combined FDR 0.85) and TNF (26 targets, 18 consistent, z 1.73, FDR 0.70) are not significant regulators: the IL-1/TNF ligands are induced, but the inference does not establish them as the principal upstream cause.
Answers	Nova-Thinking: "No — neither IL1B nor TNF achieves significant upstream-regulator overlap, despite both being predicted 'activated' ..." (*weakened*). GPT-6 Medium: "No. Despite IL1B and TNF transcript induction, neither is a significant activated upstream regulator ..." (*weakened*).
Dexamethasone in keratinocytes (GSE26487), NOVA-S3-D4-H03-S03	
Hypothesis	Dexamethasone impairs keratinocyte migration by tipping the pericellular protease/inhibitor balance toward inhibition: proteases (MMP1, MMP3, MMP9, MMP10, PLAU) are repressed while inhibitors (TIMP2 and serpins) are induced.
Step	Are tissue inhibitors of metalloproteinases induced to reinforce the protease shutdown?
Reference	Qualifies; state stays *partially supported*. TIMP2 logFC +1.02 (adjusted p 0.52), TIMP1 −0.96 (0.56), TIMP3 +0.32 (0.55): the inhibitor arm is not detected, "although absence of significance is not evidence of absence".
Answers	GPT-6 High: "No. TIMP2 rises by +1.018 but is not significant (adj p=0.520) ... The proposed shift toward protease inhibition therefore is not supported" (*weakened*). Nova-Instant: "No. None of the four tissue inhibitors of metalloproteinases (TIMP1–TIMP4) are significantly induced ..." (*partially supported*).

Table S6. Paired workflow comparisons in credit (paper, RQ1): difference in mean credit (second minus first workflow, pp) with 95% cluster-bootstrap CI, matched-pairs rank-biserial correlation, and Holm-corrected Wilcoxon p over all items; and the same on independent questions, with strict accuracy on independent questions (exact McNemar).

	Credit, all settings				**Independent questions Strict**		
Comparison	**Δ (pp)**	**95% CI**	r_{rb}	p_{Holm}	**Δ (pp)**	p_{Holm}	p_{Holm}
Thinking vs. Inst.	+1.7	[−0.3, +3.9]	+0.12	0.325	+3.6	0.133	0.084
Deep Thinking vs. Inst.	+1.6	[−0.2, +3.6]	+0.10	0.431	+2.7	0.457	0.237
Auto vs. Inst.	+2.6	[+0.5, +4.8]	+0.17	0.064	+2.7	0.461	0.241
Deep Thinking vs. Thinking	−0.1	[−2.0, +1.7]	−0.01	0.965	−0.9	1.000	1.000
Auto vs. Thinking	+0.9	[−1.1, +2.7]	+0.07	0.965	−0.9	1.000	1.000
Auto vs. Deep Thinking	+1.0	[−1.0, +2.9]	+0.07	0.965	−0.0	1.000	1.000

Table S7. Paired workflow comparisons in strict accuracy over all 1,030 graded answers (exact McNemar) and in the reference-based 0–10 grades of the 880 open answers answered by every workflow (Wilcoxon); differences are second minus first workflow, p-values Holm-corrected over the six pairs of each measure (paper, RQ1a and RQ1c).

	Strict (pp)		**Correctness**		**Completeness**		**Reasoning**	
Comparison	Δ	p_{Holm}	Δ	p_{Holm}	Δ	p_{Holm}	Δ	p_{Holm}
Thinking vs. Inst.	+3.0	0.189	+0.21	0.002	+0.50	<0.001	+0.38	<0.001
Deep Thinking vs. Inst.	+2.8	0.256	+0.25	0.001	+0.63	<0.001	+0.43	<0.001
Auto vs. Inst.	+4.3	0.026	+0.22	0.001	+0.51	<0.001	+0.39	<0.001
Deep vs. Thinking	−0.2	1.000	+0.04	1.000	+0.13	0.192	+0.04	1.000
Auto vs. Thinking	+1.3	1.000	+0.01	1.000	+0.01	0.797	+0.01	1.000
Auto vs. Deep	+1.5	1.000	−0.03	1.000	−0.11	0.294	−0.03	1.000

Table S8. Credit of the four workflows by item type, with the gap between the best and worst workflow (pp) and the Friedman p-value (paper, RQ1a–b and RQ3).

Items	n	**Inst.**	**Think.**	**Deep**	**Auto**	**Gap**	**Friedman**
By setting							
Independent questions	400	80.6	**84.2**	83.3	83.3	3.6	0.019
Conversation turns	300	75.7	74.5	75.8	**77.8**	3.3	0.360
Hypothesis steps	330	78.2	80.3	79.8	**81.1**	2.9	0.113
Independent questions, by question form							
Closed-form	143	81.1	**83.4**	79.6	81.4	3.8	0.665
Evidence integration	93	80.3	82.9	**85.8**	84.3	5.5	0.174
Short answer	86	76.2	79.6	**86.7**	81.2	10.6	0.006
Mechanistic interpretation	78	83.1	**87.4**	81.4	85.7	6.0	0.102
Independent questions, by evidence breadth							
One evidence layer	107	75.8	**76.7**	70.3	72.4	6.4	0.408
Two layers	146	84.9	88.4	**90.3**	87.4	5.4	0.010
Three or more layers	147	79.2	84.2	82.2	**84.4**	5.2	0.311
Conversation, by turn type							
Standard turns	197	74.0	72.1	75.4	**77.9**	5.8	0.165
Challenge turns	103	**78.9**	78.1	77.0	77.5	1.9	0.884
Hypothesis evaluation, judgements							
Evidence relation (per step)	300	**58.0**	53.3	52.3	56.0	5.7	0.081
Hypothesis state (per step)	300	52.3	**59.0**	50.0	57.0	9.0	0.037
Final decision	30	50.0	**63.3**	40.0	56.7	23.3	0.068

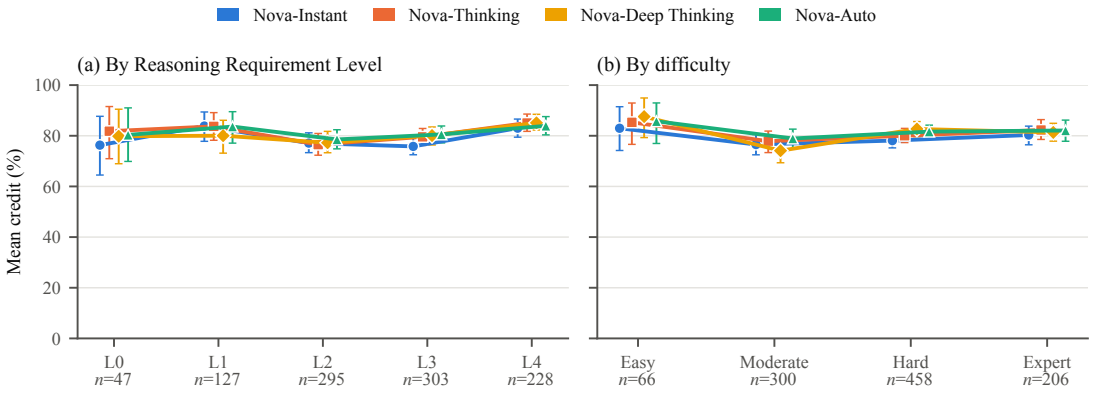


Fig. S2. Mean credit of the four workflows, all settings pooled, by (a) Reasoning Requirement Level and (b) difficulty; n: items per workflow. Credit hardly changes with the level, and no workflow-by-level interaction is significant (paper, RQ1a). Error bars: 95% cluster-bootstrap confidence intervals.

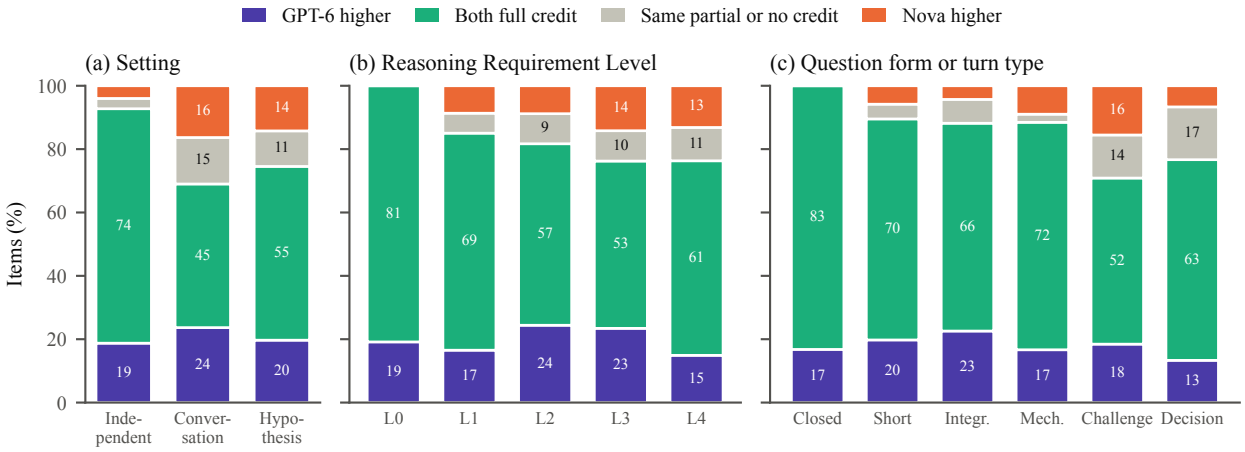


Fig. S3. Item-level agreement between Nova-Thinking and GPT-6 Medium by (a) setting, (b) Reasoning Requirement Level, and (c) question form or turn type: share of items on which GPT-6 earned more credit, both earned full credit, both earned the same partial or no credit, or Nova earned more (paper, RQ2).

Table S9. Item-level agreement between the best configuration of each pipeline (Nova-Thinking and GPT-6 Medium) on all 1,030 graded answers (paper, RQ2). Same credit: both earned full credit (full) or the same partial or no credit (lower); more credit: one pipeline earned more. Δ: GPT-6 minus Nova (pp) with a paired Wilcoxon p; 12 pairs: range of Δ over all Nova–GPT-6 configuration pairs.

		Same credit (%)		**More credit (%)**		**GPT-6 minus Nova (pp)**		
Items	n	**Full**	**Lower**	**GPT-6**	**Nova**	Δ	p	**12 pairs**
All items	1030	59.5	9.1	20.5	10.9	+7.9	<0.001	[+5.4, +10.6]
By setting								
Independent questions	400	74.0	3.2	18.8	4.0	+11.6	<0.001	[+9.6, +15.3]
Conversation turns	300	45.3	14.7	23.7	16.3	+6.5	0.002	[+1.0, +11.0]
Hypothesis items	330	54.8	11.2	19.7	14.2	+4.5	0.018	[+3.2, +6.7]
By Reasoning Requirement Level								
L0	47	80.9	0.0	19.1	0.0	+18.2	0.004	[+15.8, +23.7]
L1	127	68.5	6.3	16.5	8.7	+7.5	0.007	[+5.1, +11.4]
L2	295	57.3	9.5	24.4	8.8	+11.9	<0.001	[+6.7, +11.9]
L3	303	52.8	9.6	23.4	14.2	+6.0	<0.001	[+5.0, +14.7]
L4	228	61.4	10.5	14.9	13.2	+2.2	0.501	[+1.2, +4.3]
By difficulty								
Easy	66	74.2	1.5	16.7	7.6	+10.3	0.015	[+8.1, +14.1]
Moderate	300	62.0	8.0	22.3	7.7	+11.9	<0.001	[+7.2, +15.6]
Hard	458	56.1	9.2	21.4	13.3	+6.8	<0.001	[+3.6, +11.9]
Expert	206	58.7	13.1	17.0	11.2	+3.5	0.064	[+1.8, +5.5]
By question form or turn type								
Closed-form	143	83.2	0.0	16.8	0.0	+16.6	<0.001	[+12.3, +20.4]
Short answer	86	69.8	4.7	19.8	5.8	+12.9	0.004	[+1.9, +18.5]
Evidence integration	93	65.6	7.5	22.6	4.3	+10.5	<0.001	[+4.7, +13.1]
Mechanistic interpretation	78	71.8	2.6	16.7	9.0	+6.1	0.115	[+6.1, +15.3]
Standard turns	197	41.6	15.2	26.4	16.8	+8.6	0.003	[+0.6, +13.4]
Challenge turns	103	52.4	13.6	18.4	15.5	+4.3	0.311	[+2.0, +9.6]
Hypothesis steps	300	54.0	10.7	20.3	15.0	+4.5	0.029	[+3.7, +6.8]
Final decisions	30	63.3	16.7	13.3	6.7	+5.0	0.317	[−3.3, +15.0]
By evidence breadth (all settings)								
1 layer	262	64.1	6.9	20.6	8.4	+11.6	<0.001	[+9.2, +15.5]
2 layers	392	58.9	10.7	19.4	11.0	+7.3	<0.001	[+4.8, +9.2]
3+ layers	376	56.9	9.0	21.5	12.5	+6.0	<0.001	[+4.9, +9.5]
By item provenance								
Manually curated	312	55.8	9.0	21.8	13.5	+6.6	<0.001	[+1.5, +7.1]
LLM-assisted	718	61.1	9.2	19.9	9.7	+8.4	<0.001	[+6.7, +12.2]

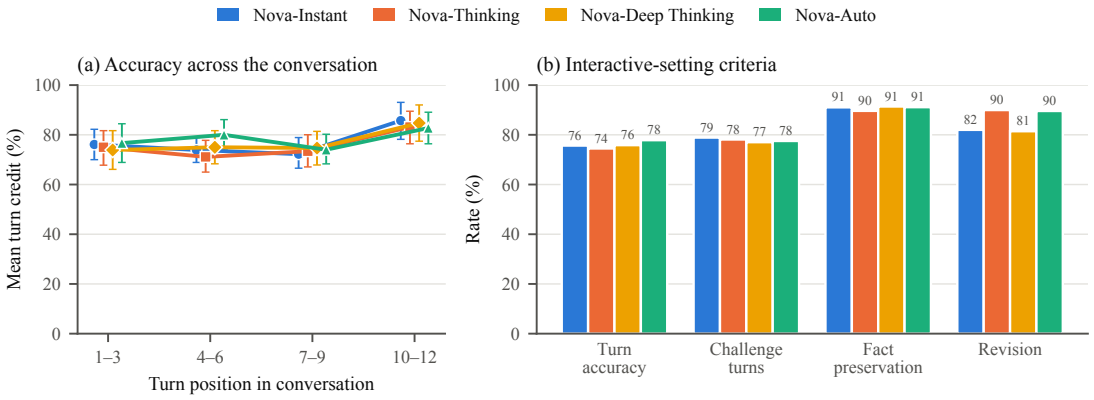


Fig. S4. Multi-turn conversations, four Nova workflows (paper, RQ3). (a) Mean turn credit by position in the conversation. (b) Turn accuracy, accuracy on challenge turns, preservation of earlier facts, and revision where the reference annotations expect it.

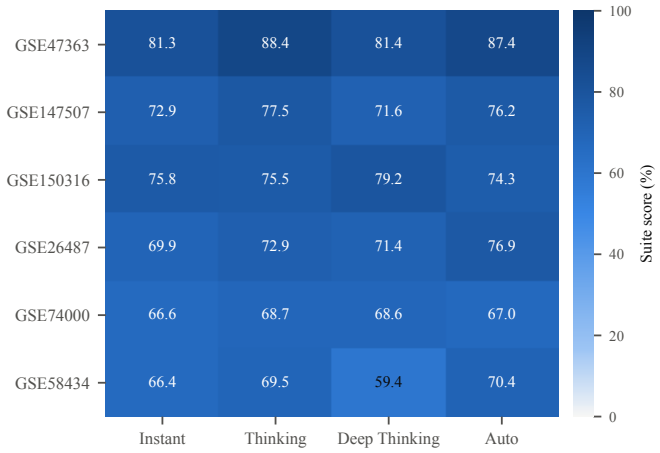


Fig. S5. Suite score by dataset and workflow: some datasets are harder for every workflow, which drives the correlation between independent-question and conversation accuracy across workflow–dataset combinations (paper, RQ3).

Table S10. The four Nova workflows on the same 4,120 answers graded by Claude Opus 5.5 and by GPT-6 (paper, RQ4); bold: best workflow under each grader (lowest for unsupported claims). Strict accuracy here is on independent questions.

	Graded by Claude Opus 5.5				Graded by GPT-6			
Metric	**Inst.**	**Think.**	**Deep**	**Auto**	**Inst.**	**Think.**	**Deep**	**Auto**
Suite score (%)	72.1	**75.4**	71.9	75.4	66.8	**72.9**	69.1	72.1
Independent-question accuracy (%)	80.6	**84.2**	83.3	83.3	74.5	**77.9**	77.8	76.7
Strict accuracy (full match) (%)	72.7	**77.9**	77.2	76.9	62.7	69.2	**69.9**	68.0
Conversation turn accuracy (%)	75.7	74.5	75.8	**77.8**	73.7	79.0	79.2	**79.8**
Hypothesis score (%)	60.2	**67.5**	56.6	64.9	52.2	**61.9**	50.3	59.8
Step credit (%)	78.2	80.3	79.8	**81.1**	70.2	**75.9**	74.2	75.2
Evidence relation (%)	**58.0**	53.3	52.3	56.0	**34.7**	32.0	28.7	32.7
Hypothesis state (%)	52.3	**59.0**	50.0	57.0	39.7	46.3	36.7	**47.7**
Final decision (%)	50.0	**63.3**	40.0	56.7	46.7	**63.3**	40.0	56.7
Open independent questions questions (credit, %)								
Short answer	76.2	79.6	**86.7**	81.2	77.2	**80.3**	76.9	77.7
Evidence integration	80.3	82.9	**85.8**	84.3	63.8	70.1	**72.6**	72.0
Mechanistic interpretation	83.1	**87.4**	81.4	85.7	75.5	78.0	**82.1**	75.2
Grades								
Completeness (0–10)	6.81	7.31	**7.44**	7.33	6.39	6.82	**6.86**	6.79
Reasoning (0–10)	7.29	7.67	**7.71**	7.68	6.42	6.82	**6.84**	6.79
Unsupported claims (%)	12.4	10.8	**10.4**	10.7	**10.3**	11.0	12.0	12.3

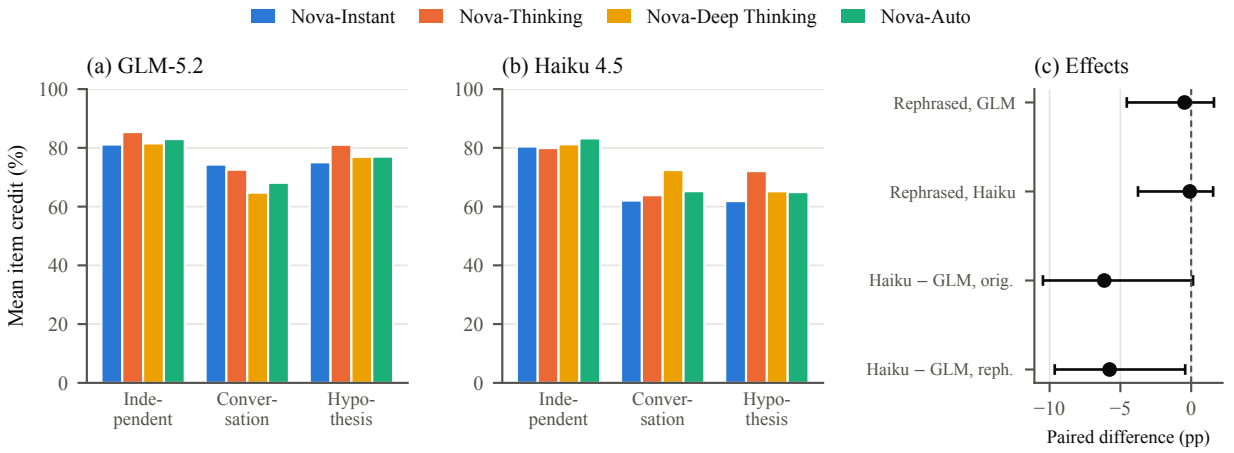


Fig. S6. Robustness subset (paper, RQ4 and Table 7). (a, b) Mean credit by setting and workflow for GLM-5.2 and Claude Haiku 4.5, original and rephrased wording pooled: for Haiku 4.5, the best workflow depends on the setting. (c) Paired differences in credit with 95% cluster-bootstrap confidence intervals, as in Fig. 4c.

Table S11. Grader agreement on the same 3,542 open answers (paper, RQ4): mean credit under each grader, the difference (GPT-6 minus Claude), the share of answers given the same credit, the share GPT-6 graded lower, and linearly weighted Cohen's κ.

		Mean credit (%)			Same answers, two graders		
Open answers	n	**Claude**	**GPT-6**	Δ **(pp)**	**Same credit (%)**	**GPT-6 lower (%)**	κ_w
All open answers	3,542	79.8	75.5	−4.3	56.1	25.2	0.20
By setting							
Independent questions	1,028	83.8	74.3	−9.5	57.0	28.8	0.22
Conversation turns	1,198	76.1	78.0	+2.0	55.8	20.3	0.22
Hypothesis items	1,316	80.1	74.1	−6.0	55.7	26.8	0.17
By Reasoning Requirement Level							
L0–L1	324	84.1	77.9	−6.2	60.5	24.4	0.18
L2	981	75.3	76.8	+1.4	55.0	21.7	0.18
L3	1,210	78.7	77.2	−1.5	56.6	22.7	0.28
L4	907	84.6	72.2	−12.4	54.0	32.2	0.13
By difficulty							
Easy	108	94.4	86.1	−8.3	76.9	19.4	0.38
Moderate	881	75.7	76.8	+1.1	55.2	21.8	0.18
Hard	1,730	80.4	75.8	−4.6	56.9	24.7	0.22
Expert	823	81.1	72.1	−9.1	52.9	30.5	0.16
By question form or turn type							
Short answer	344	83.4	79.9	−3.5	64.2	20.9	0.24
Evidence integration	372	83.3	67.9	−15.5	50.8	36.0	0.20
Mechanistic interpretation	312	84.8	75.8	−9.0	56.4	28.8	0.24
Standard turns	787	74.7	79.2	+4.5	56.9	17.7	0.27
Challenge turns	411	78.7	75.8	−2.9	53.8	25.3	0.13
Hypothesis steps	1,196	80.1	74.9	−5.2	54.8	26.8	0.16
By workflow (answer source)							
Nova-Instant	887	78.0	71.6	−6.3	52.3	29.1	0.20
Nova-Thinking	887	79.7	76.8	−2.9	57.5	23.2	0.24
Nova-Deep Thinking	881	80.6	77.1	−3.6	57.3	23.7	0.17
Nova-Auto	887	81.0	76.5	−4.5	57.4	24.7	0.19

Question (GSE47363; multiple choice, L2, hard). *Which regulator is significant by target over-representation yet has an undetermined predicted direction (equal activating and inhibiting targets) and an essentially unchanged own transcript?* A. PAK2 B. TP53 C. SREBF2 D. BDNF. **Reference:** A.

Nova-Deep Thinking, correct (judge 5.8/10): "**PAK2**: logFC = 0.110 (essentially unchanged); 3 of 6 DE targets activation-consistent, so the direction is undetermined; uncorrected $p = 0.0052$. **TP53**: 48 of 55 targets activation-consistent, so the direction is determined. ... PAK2 is the only regulator matching all three criteria." Nova-Auto answered A in the same way.

Nova-Instant, wrong (C; judge 3.0/10, 40% unsupported claims): "... the predicted direction is 'inhibited.' Wait — the question specifies an *undetermined* predicted direction. ... The best fit is **SREBF2**."

Nova-Thinking, wrong (B; judge 3.2/10, 55% unsupported claims): "... TP53 is the only regulator that is significant by ORA ..."

Fig. S7. The same evidence, applied differently: answers of the GLM-5.2 workflows to one independent question (robustness subset, original wording), quoted verbatim and shortened (...). PAK2 is significant by target over-representation (FDR 0.0021) with an activation z-score of 0 (3 of 6 DE targets in each direction) and an unchanged transcript (logFC 0.11, adjusted p 0.67). Paper, Discussion.

Table S12. Reference-based and reference-free grading of the same 4,120 answers under each grader (paper, RQ4). Ground truth: exact-match grades of closed-form answers.

Measure	**Claude Opus 5.5**	**GPT-6**
Within a grader: reference-based against reference-free score		
Spearman ρ, all answers	0.39	0.13
Spearman ρ, open answers	0.36	0.19
Within one point, open answers (%)	56.2	40.0
Reference-free minus reference-based, open answers	+0.61	+1.39
Reference-free score against ground truth (closed-form answers, exact match)		
AUC, correct against incorrect	0.91	0.56
Mean score, correct / incorrect answers	8.53 / 5.12	8.10 / 7.69
Unsupported-claim estimate against score (Spearman ρ, open answers)		
Reference-free score	−0.63	−0.70
Reference-based score	−0.16	−0.16
Between graders (Spearman ρ, all / open answers)		
Reference-based scores	0.47 / 0.19	
Reference-free scores	0.07 / 0.05	

References

1. Draghici, S., Khatri, P., Tarca, A.L., Amin, K., Done, A., Voichiţa, C., Georgescu, C., Romero, R.: A systems biology approach for pathway level analysis. Genome Research **17**(10), 1537–1545 (2007)
2. Wang, Y., Huang, J.W., Castella, M., Huntsman, D.G., Taniguchi, T.: p53 is positively regulated by mir-542-3p. Cancer Research **74**(12), 3218–3227 (2014)
3. Blanco-Melo, D., Nilsson-Payant, B.E., Liu, W.C., Uhl, S., Hoagland, D., Møller, R., Jordan, T.X., Oishi, K., Panis, M., Sachs, D., Wang, T.T., Schwartz, R.E., Lim, J.K., Albrecht, R.A., tenOever, B.R.: Imbalanced Host Response to SARS-CoV-2 Drives Development of COVID-19. Cell **181**(5), 1036–1045 (2020). `https://doi.org/10.1016/j.cell.2020.04.026`
4. Desai, N., Neyaz, A., Szabolcs, A., Shih, A.R., Chen, J.H., Thapar, V., Nieman, L.T., Solovyov, A., Mehta, A., Lieb, D.J., et al.: Temporal and Spatial Heterogeneity of Host Response to SARS-CoV-2 Pulmonary Infection. medRxiv **11**(1), 6319 (2020)
5. Stojadinovic, O., Lee, B., Vouthounis, C., Vukelic, S., Pastar, I., Blumenberg, M., Brem, H., Tomic-Canic, M.: Novel genomic effects of glucocorticoids in epidermal keratinocytes: inhibition of apoptosis, interferon-γ pathway, and wound healing along with promotion of terminal differentiation. Journal of Biological Chemistry **282**(6), 4021–4034 (2007)
6. Rodrigues, R.M., Govaere, O., Roskams, T., Vanhaecke, T., Rogiers, V., De Kock, J.: Gene expression data from acetaminophen-induced toxicity in human hepatic in vitro systems and clinical liver samples. Data in Brief **7**, 1052–1057 (2016)
7. Himes, B.E., Koziol-White, C., Johnson, M., Nikolos, C., Jester, W., Klanderman, B., Litonjua, A.A., Tantisira, K.G., Truskowski, K., MacDonald, K., et al.: Vitamin d modulates expression of the airway smooth muscle transcriptome in fatal asthma. PLoS One **10**(7), e0134057 (2015)